



IIT, Kanpur

ISTVT: Interpretable Spatial-Temporal Video Transformer for Deepfake Detection

Course Project **EE656**

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Why ISTVT?

Deepfake Threat: Rapid growth of Deepfake technology poses serious risks to information security and personal privacy.



Current Methods

Existing spatial-temporal models like RNNs and 3D CNNs have limitations in both performance and interpretability.



Proposed Solution: ISTVT

- Uses decomposed spatial-temporal self-attention to isolate spatial artifacts and temporal inconsistencies.

- Introduces a self-subtract mechanism to enhance feature discrimination.

Interpretability- Uses relevance propagation to visualize important spatial and temporal regions.



Results:

- Outperforms on intra- and cross-dataset Deepfake detection using datasets like FaceForensics++, DFDC, CelebDF, etc.

- Offers both accuracy and transparency, improving trust in Deepfake detection models.

Index Terms

01 Spatial

02 Temporal

03 Xception

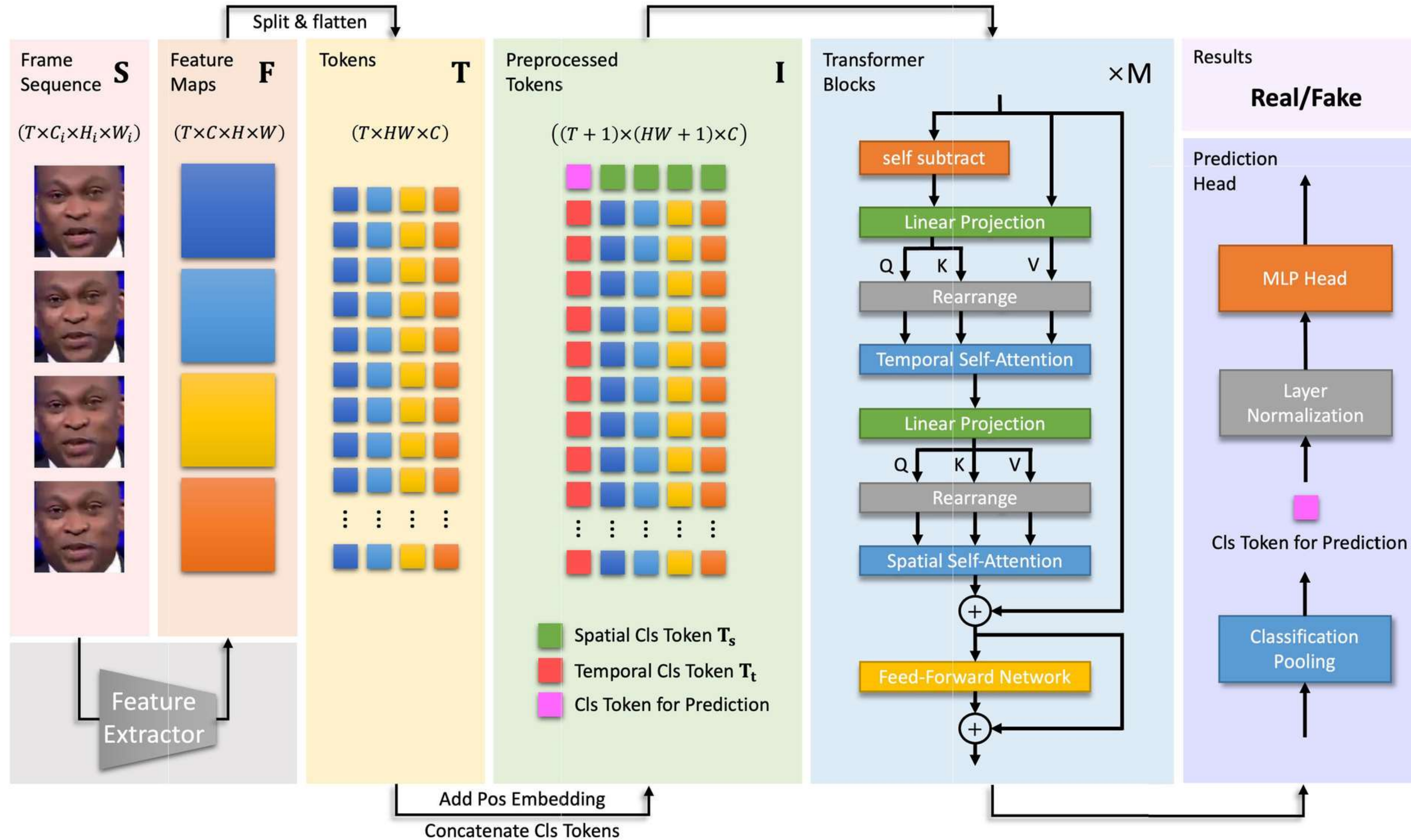
04 Transformers

05 Self-attention

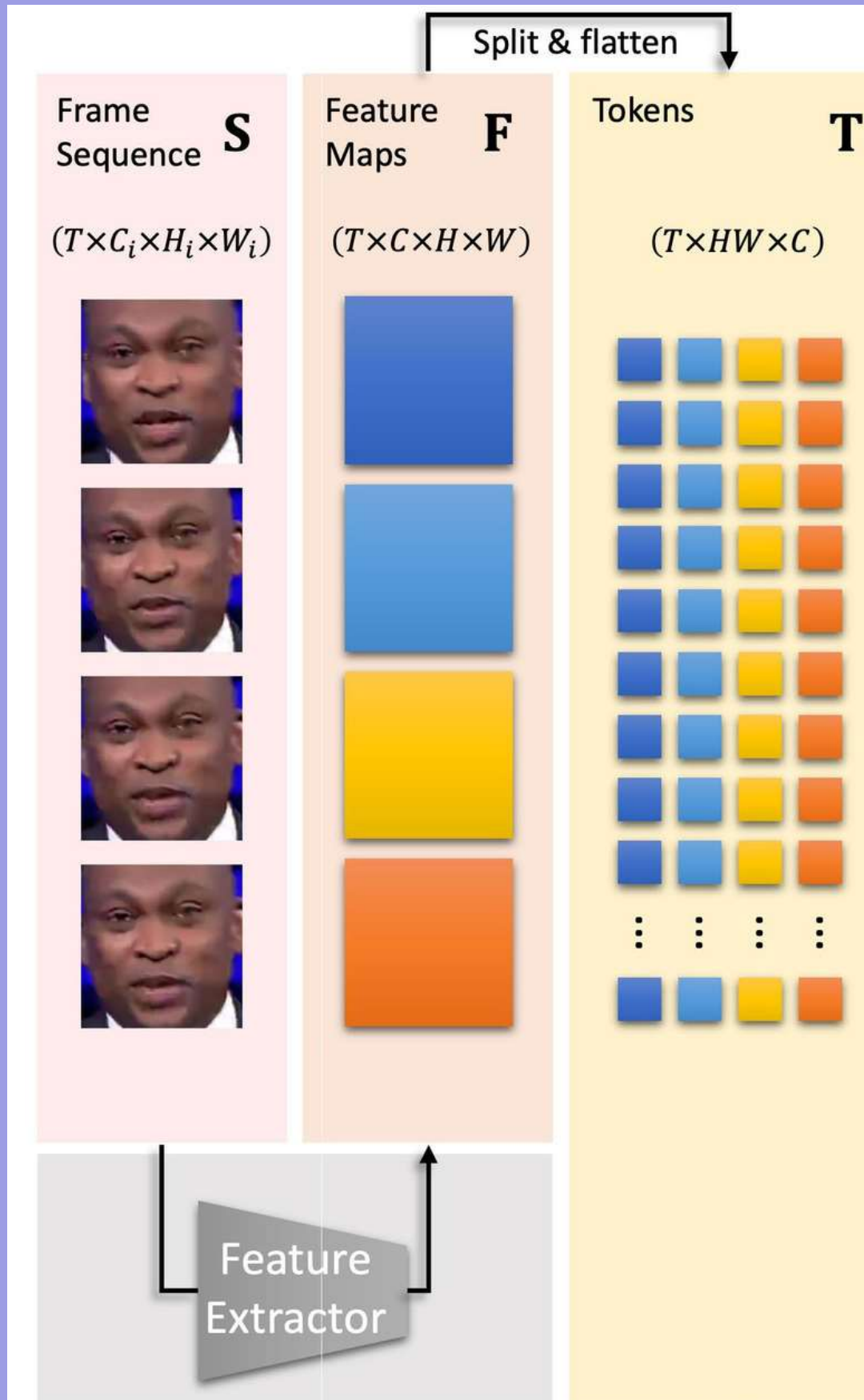
06 Self-subtraction

07 Layer-wise relevance
Propagation

Structure of ISTVT



Methodology



1. Input and feature extraction

Given a video $V = \{F_1, F_2, \dots, F_T\}$ of T frames,

Each frame is processed by Xception, and the combined

The feature sequence is:

$$X = \{X_1, X_2, \dots, X_T\} \in \mathbb{R}^{T \times H \times W \times C}$$

Where:

- T : number of frames (temporal dimension)
- H, W : spatial height and width of feature maps
- C : number of channels

The feature maps F are then split into 1×1 patches and flattened into vectors, each vector represents C feature patterns known as 'TOKENS'

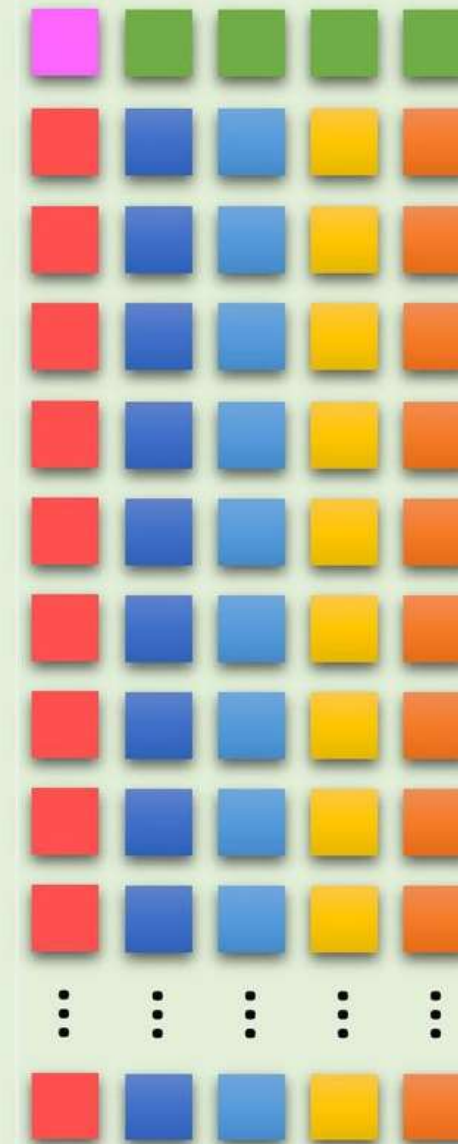
2. Adding more dimensions

- Tokens $T \in \mathbb{R}^{T \times HW \times C}$ are given.
- Concatenate with spatial classification tokens $T_s \in \mathbb{R}^{T \times 1 \times C}$
- Result becomes $T \in \mathbb{R}^{T \times (HW+1) \times C}$.
- Concatenate with temporal classification tokens $T_t \in \mathbb{R}^{1 \times (HW+1) \times C}$.
- Final shape becomes $(T+1) \times (HW+1) \times C$.

Preprocessed
Tokens

I

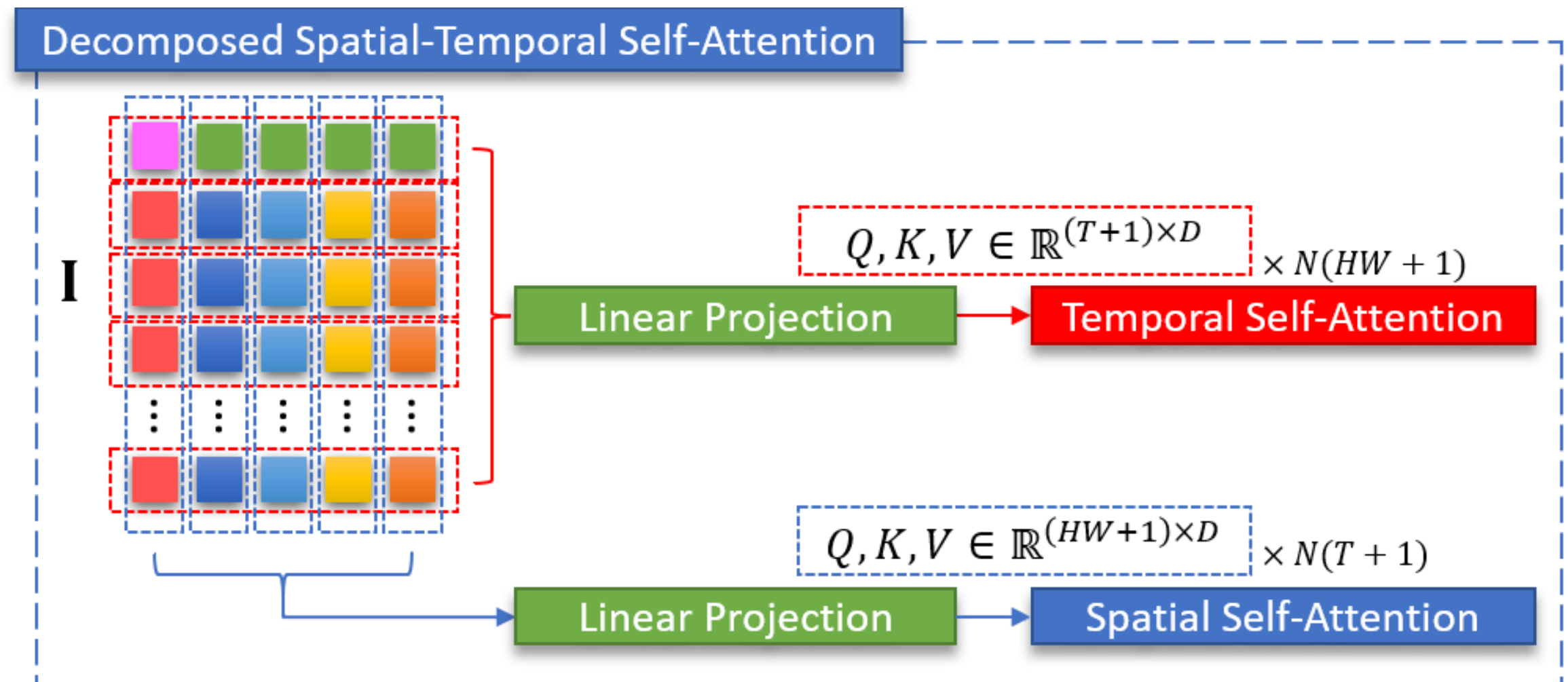
$$((T + 1) \times (HW + 1) \times C)$$



-  Spatial Cls Token T_s
-  Temporal Cls Token T_t
-  Cls Token for Prediction

3. Self-Attention

- Now the tokens are passed through M transformer Heads.
- In each head, the featured maps are updated in parallel using the concept of 'SELF-ATTENTION'.
- For eg, we then conduct self-attention on the spatial and temporal dimensions respectively.



Q, K, V are linear projections of input tokens.
Compute $(\text{SCORE})_{ij} = (Q_i) \cdot (K_j)^T$
Scaled score = $(\text{SCORE})_{ij} / (d_k)$
Apply softmax $\rightarrow$ turns scores into weights that sum to 1.

4. Self-Subtract

- The self-subtract mechanism helps ISTVT ignore repeated/static content and focus on **what's changing between frames**, which is useful for spotting Deepfake signs.
- *Temporal self-attention uses subtracted tokens to create Q and K , capturing changes between frames, while V comes from original tokens. This helps the model focus on motion and detect temporal inconsistencies.*

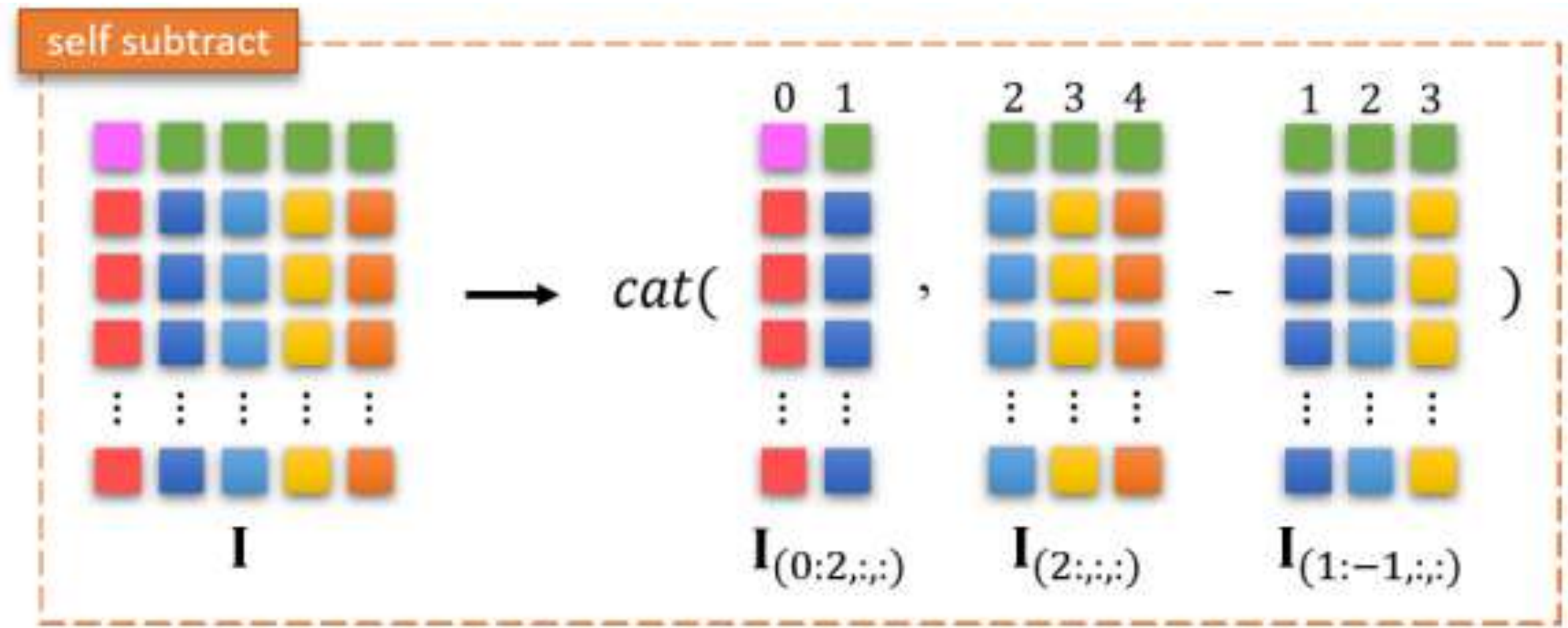


Fig. 4. Details of the proposed self-subtract mechanism.

Experiments

(1) Intra-Dataset Comparisons

ISTVT is tested against strong models across multiple datasets and shows superior accuracy and robust generalization. It outperforms CNNs and older transformers, especially due to its focus on temporal and spatial inconsistencies, which are key in Deepfake detection.

(2) Cross-Dataset Comparisons

ISTVT shows strong generalization to unseen datasets, especially complex ones like DFDC. It outperforms frame-based methods by using short-term temporal inconsistencies through separate spatial-temporal attention. Even against powerful models like FTCN, ISTVT holds advantages in challenging real-world conditions.

Robustness to Perturbations

We introduce three common noises and perturbations, i.e., JPEG compression, downscale, and random dropout, and apply them respectively to the testing set.

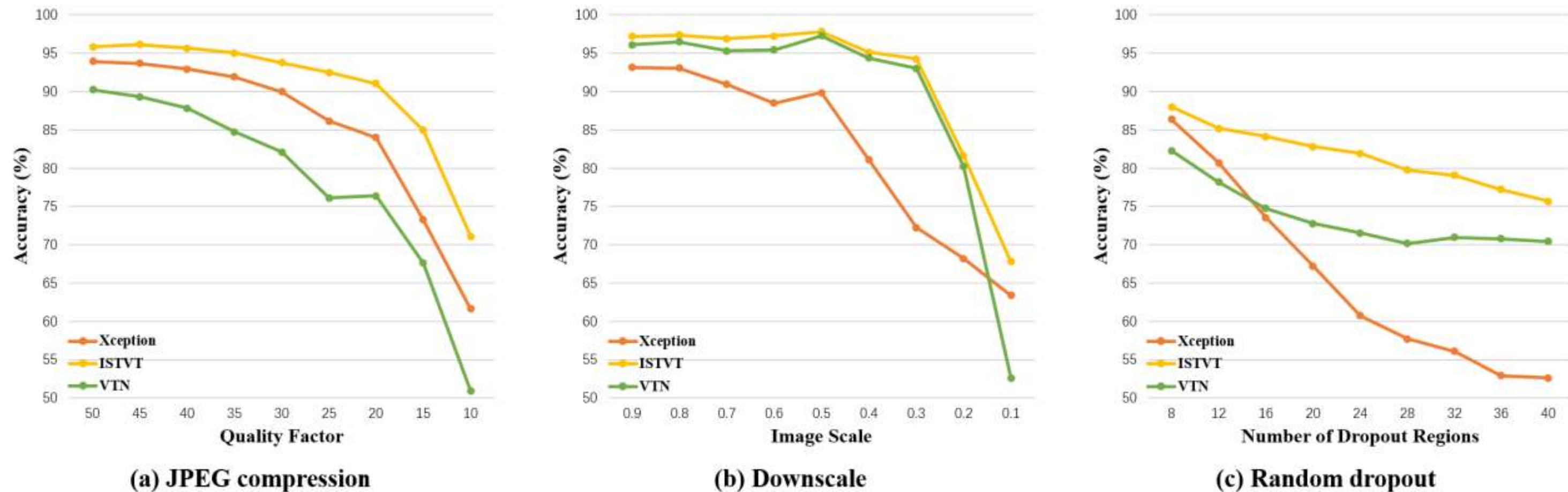


Fig. 6. Robustness test of ISTVT and two baseline methods.

ISTVT shows strong robustness to noise like compression, downscaling, and dropout. Its design — especially self-subtract and separated spatial-temporal attention — makes it better than both VTN and Xception in handling real-world video distortions.

Different attention structures

- The results reported in Table III demonstrate that among all decomposed attention mechanisms, especially with temporal attention first, give the best performance.
- Generally, the Self-subtract mechanism improves performance across all self-attention variants.

TABLE III

DEEPPAKE DETECTION ACCURACY (%) AND AUROC (%) OF ISTVT WITH DIFFERENT SELF-ATTENTION VARIANTS. TEMPORAL FIRST AND SPATIAL FIRST INDICATE THE DECOMPOSED ATTENTION WITH TEMPORAL OR SPATIAL FIRST

Methods	FF++		Celeb-DF	
	Acc	AUROC	Acc	AUROC
Vanilla	93.3	94.0	99.0	99.5
Temporal First	94.9	96.3	99.6	99.8
+ Self-subtract	96.0	96.7	99.8	99.9
Spatial First	93.4	95.8	99.6	99.6
+ Self-subtract	93.8	96.0	99.6	99.8
Parallel Attention	90.9	92.6	98.8	99.3
+ Self-subtract	91.7	93.2	99.0	99.5
Decomposed Block	78.2	80.5	98.3	99.1
+ Self-subtract	79.6	81.9	98.7	99.2
Spatial Only	92.8	93.5	98.5	99.0
Temporal Only	92.5	93.3	98.0	98.4
+ Self-subtract	92.7	93.8	98.3	98.9

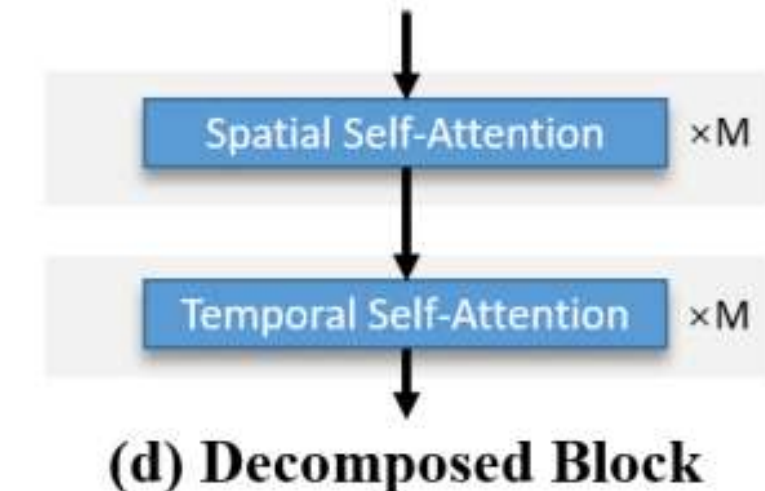
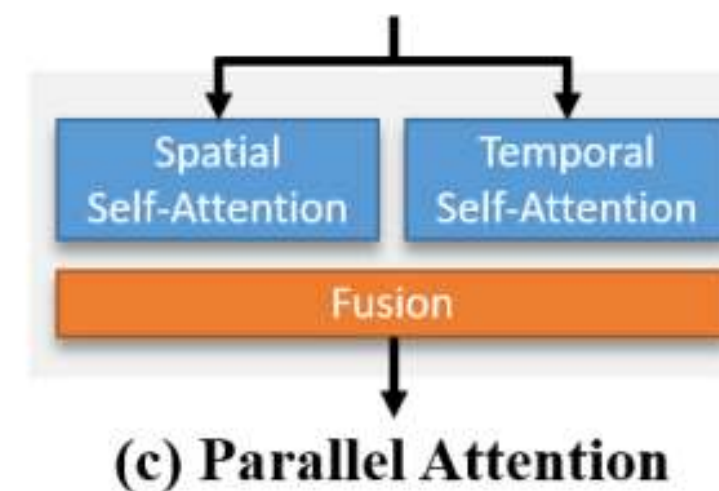
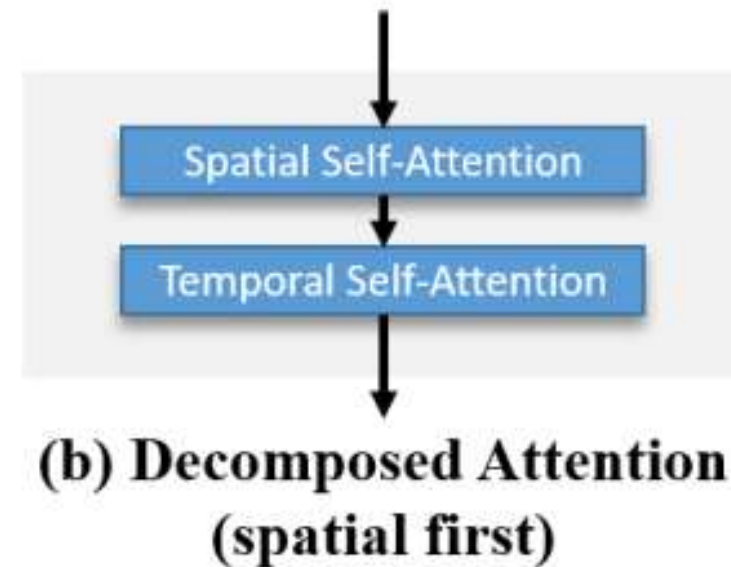
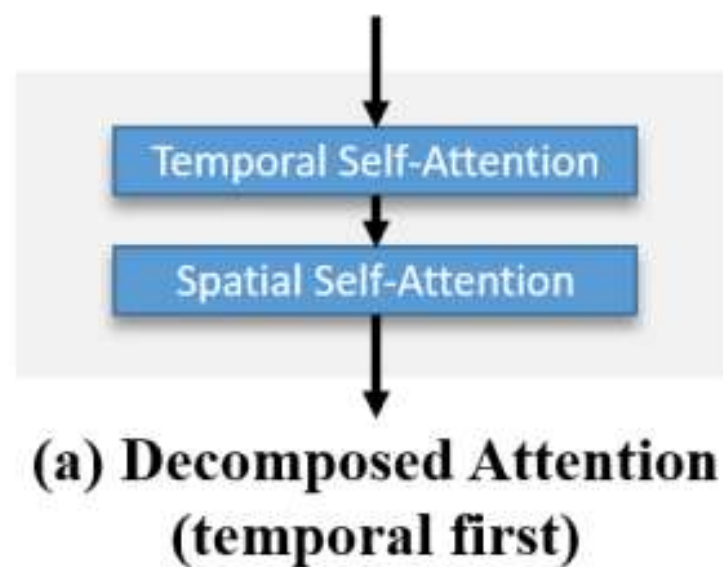


Fig. 7. Four variants of the spatial-temporal self-attention.

Transformer Settings

- On varying the ‘hyperparameter’ settings of the video transformer, we observed that results mainly depend on “M” and “T”.
- For ‘T’, using a small number of frames (like T=2) is enough for deepfake detection, and further increasing accuracy increases till a certain limit.

TABLE IV

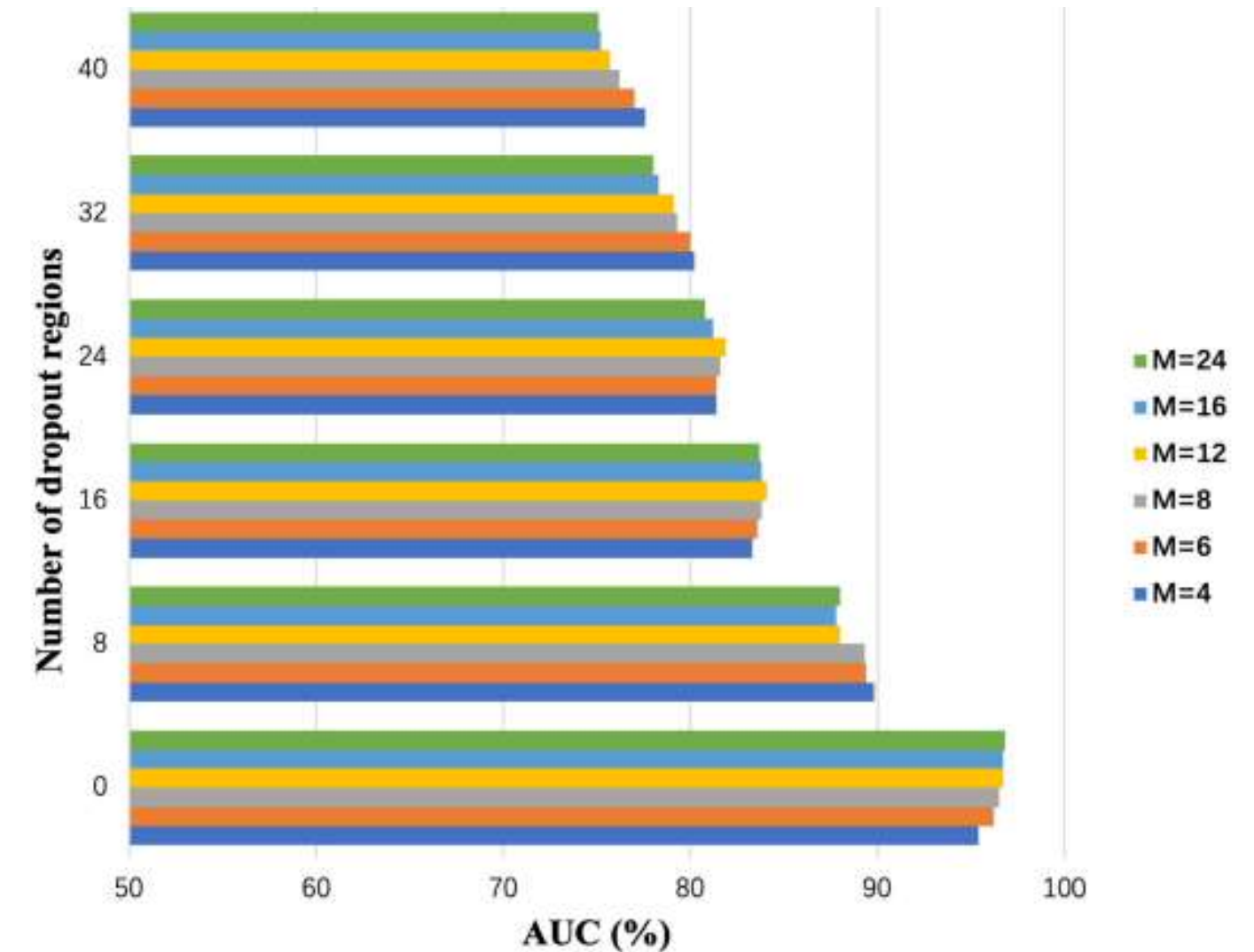
COMPARISON BETWEEN DIFFERENT SEQUENCE LENGTH T

T	2	4	6	8	12	16
AUC (%)	96.1	96.5	96.7	96.7	96.6	96.6
GFLOPs	99.6	167.9	236.1	304.4	440.9	577.4

TABLE V

COMPARISON BETWEEN DIFFERENT MODEL DEPTH M

M	4	6	8	12	16	24
AUC (%)	95.4	96.2	96.5	96.7	96.7	96.8
GFLOPs	89.7	126.3	162.9	236.1	309.4	455.8



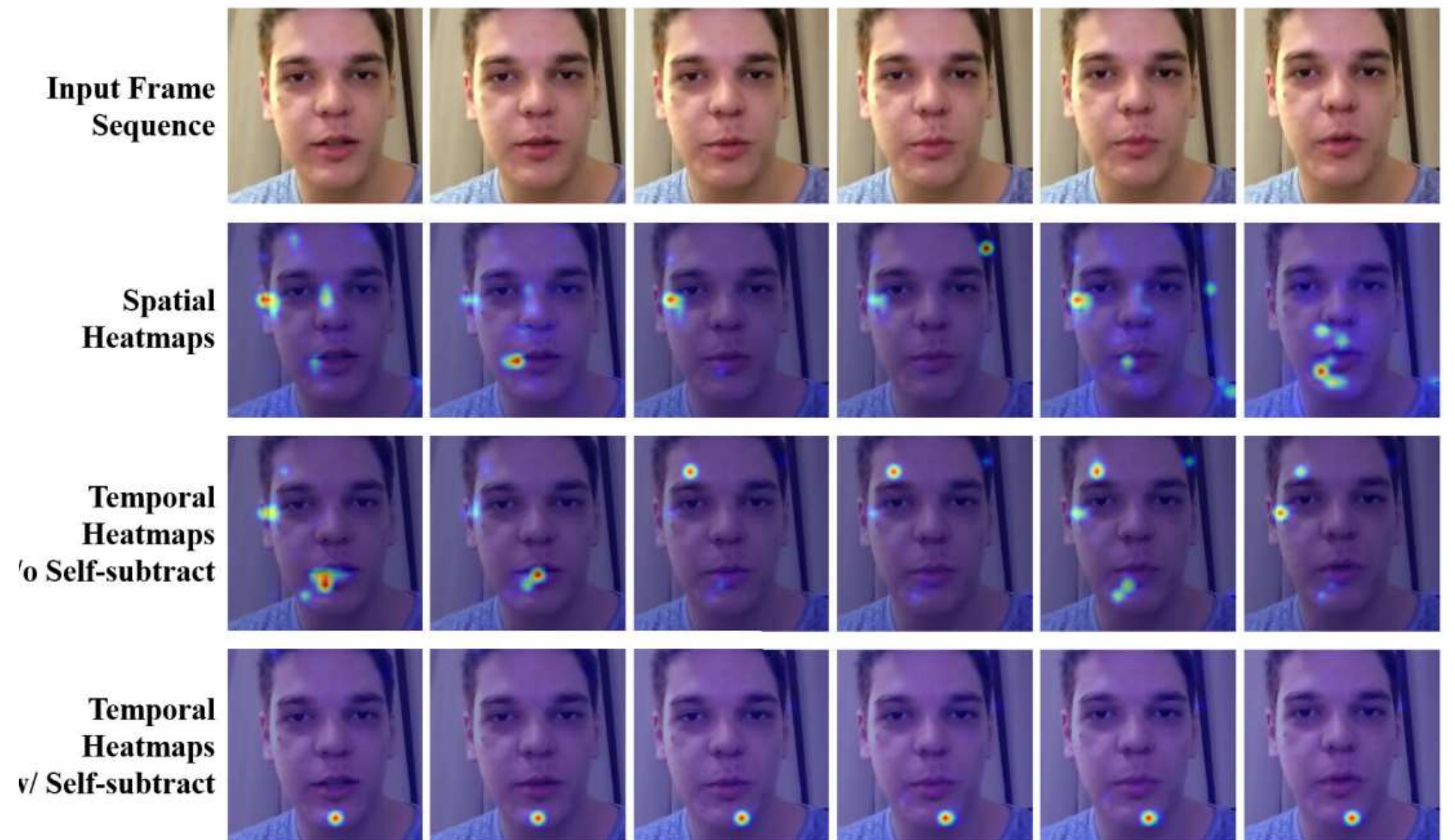
- For ‘M’, making the model deeper by taking high ‘M’ doesn't help much because deepfake clues don't lie in complex features, and it also increases the complexity.
- Using ‘6’ input frames (T=6) and ‘12’ transformer layers (M=12) gives the best balance between accuracy and computational cost.

Interpretability by VISUALIZATION

Heatmaps

A heatmap shows which image regions most influence the model's prediction — helping us interpret what the model is focusing on.

- The temporal module detects movement-related inconsistencies across frames, while the spatial module finds visual errors within each frame
- Along with self-subtract, they cover both motion and appearance flaws in deepfakes.



Example of a challenging case. The input frame sequence is sampled from a real video in FaceForensics++ dataset, the model without self-su

Perturbation-Based Explanation Test

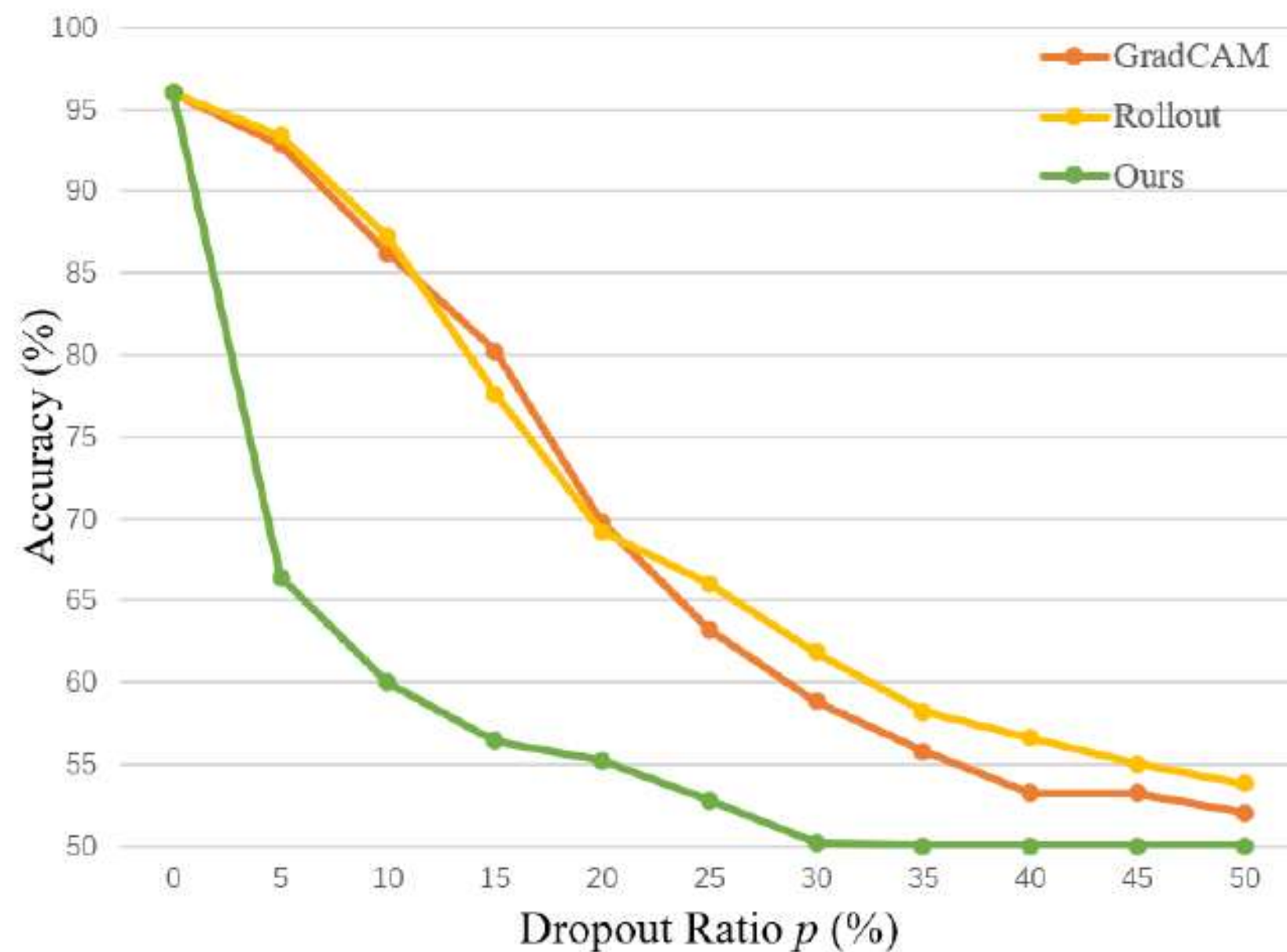


Fig. 12. Results of the quantitative test of different explanation methods.

In this test, we remove the top $p\%$ of pixels (most important ones observed with the help of heatmaps) and test how the model's accuracy changes.

On FaceForensics++, removing just 10% of ISTVT heatmap pixels drops accuracy below 60%, proving high precision.

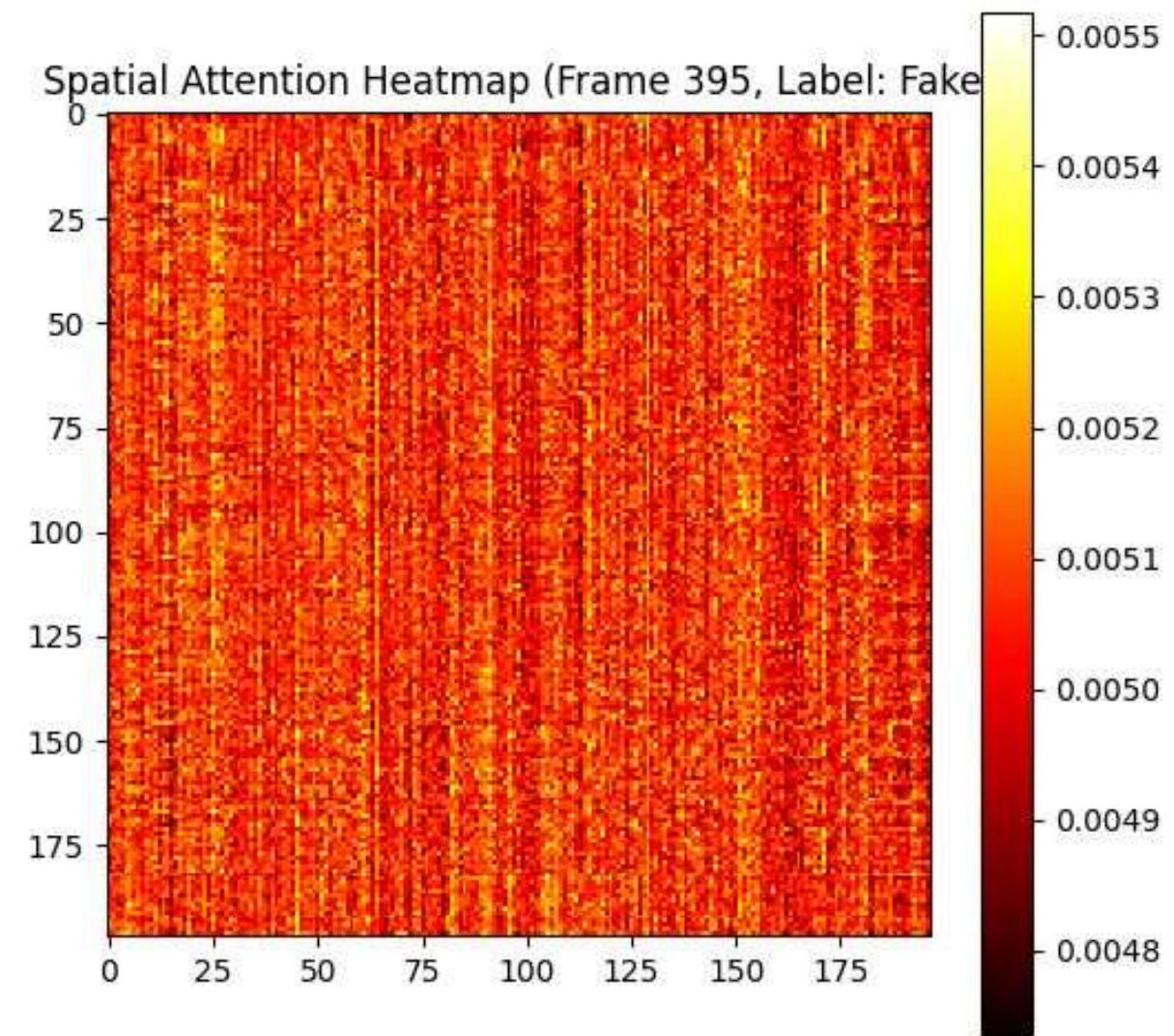
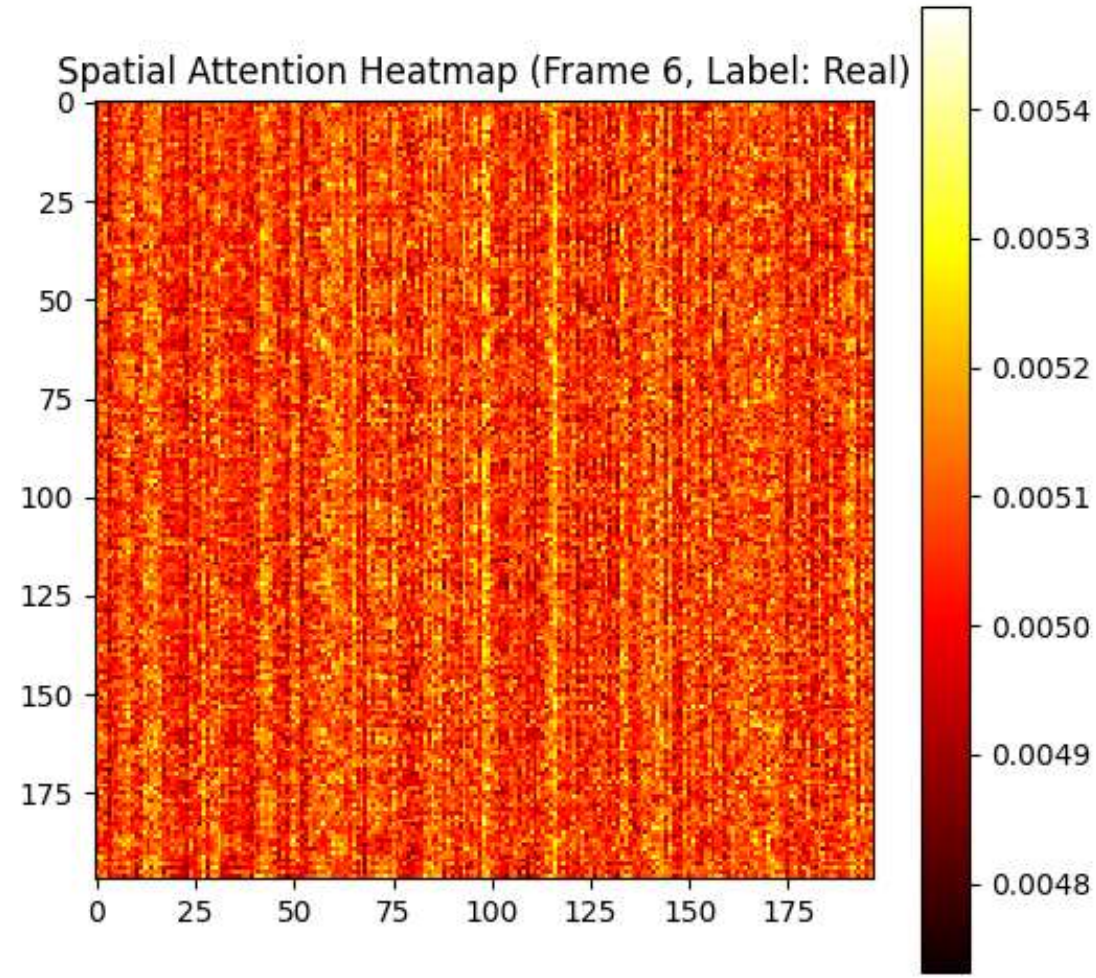
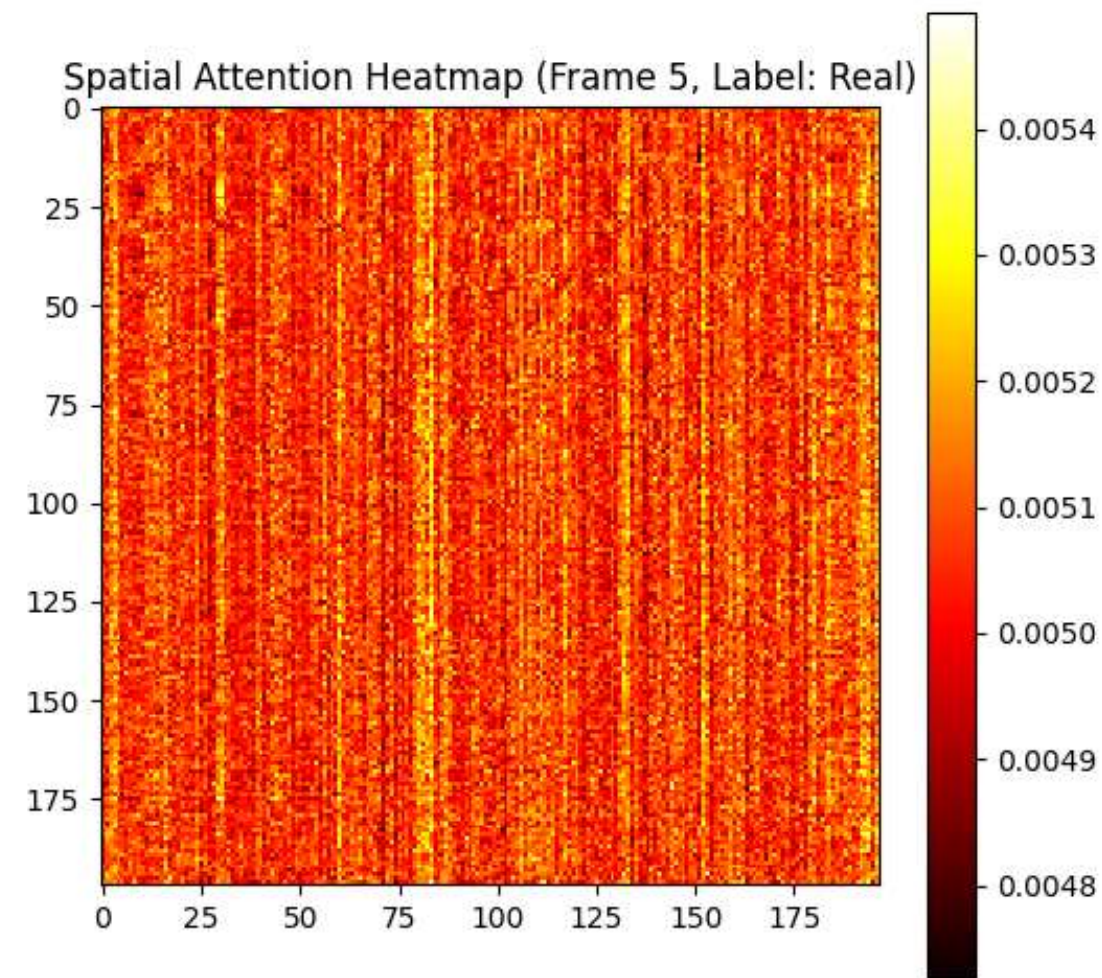
Rollout and GradCAM need over 30% removed to have similar impact, showing less accurate focus.

This proves that “ISTVT” has strong explanatory ability compared to other methods.

Novelty and Achievements of ISTVT

- Proposed ISTVT, a novel video transformer that separates spatial and temporal attention and introduces a self-subtract mechanism to better detect Deepfake artifacts.
- Developed a new interpretability method using Layer-wise Relevance Propagation (LRP) to visualize spatial and temporal attention separately—first of its kind in video transformers.
- Achieved robust performance across multiple Deepfake datasets (e.g., FaceForensics++, DFDC, Celeb-DF), with both quantitative results and visualizations validating effectiveness.

Code Implementation by Us





Thank You

09 July, 2025